Xinbao Qiao

xinbaoqiao@cuhk.edu.hk | Homepage | GitHub | Google Scholar

Education

- **The Chinese University of Hong Kong** — *2026 - present*
  PhD student in Information Engineering — Hong Kong
- **Zhejiang University** — *09/2022 - 12/2025*
  Master in Artificial Intelligence — Zhejiang
  - Relevant courses: AI Algorithms and Systems (100), Secure Artificial Intelligence (97)
  - Major GPA: 90/100 (Rank: 3/25)
- **Shandong University** — *09/2018 - 07/2022*
  Bachelor in Communication Engineering — Shandong
  - Major GPA: 82.47/100
  - Award: Third-class Academic Award 2018-2019

Research Interests

- Lifecycle management of data in AI models, focusing on theoretical methods and practical problems that arise as data are generated, used, and deleted.
- The intersection of AI with network, including AI for communication, communication for AI.

Research Experience

- **Research on Data-Centric ML Systems** — ***M.Eng. Student*, *03/2023 - 12/2025***
  Advisor: Prof. Meng Zhang, Zhejiang University
  - **Data Influence Attribution.** Developed theoretically guaranteed approaches to quantify and analyze the influence of data on learning models, and designed unlearning techniques to remove these influences.
  - **Trade-offs in Trustworthy Artificial Intelligence.** Investigated and developed data-centric methodologies to balance fairness, robustness, privacy, and utility in machine learning systems.
  - **Cloud-Edge Collaborative Human-Space Healthcare.** Architected cloud-edge collaborative frameworks for real-time multimodal data monitoring of human health using video, audio, and sensor signals.
- **Research on Trustworthy LLM systems** — ***Full-time Research Intern***, *06/2025 - 12/2025*
  Advisor: PANG Yan, James, National University of Singapore
  - **Illusory Pattern Perception in LLMs.** Examined whether LLMs exhibit illusory pattern perception, a known human cognitive bias, and offered an interpretable framework to account for LLM behavior.
  - **Synthetic Data Evaluation.** Developed Wasserstein-based methods for evaluating synthetic data under limited real-data access, with emphasis on distributed evidence aggregation and low-resource validation.

Skills

- **Technical Skills:** Python (advanced), MATLAB (advanced), LaTeX (advanced), Markdown (advanced), C/C++, Linux, Java, and Frameworks & Tools: {TensorFlow, PyTorch, SQL, Git, Linux, Docker}, etc.
- **Language:** Mandarin Chinese (native), English

PUBLICATIONS

[1] When Sample Selection Bias Precipitates Model Collapse
Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang.
Forty-Third International Conference on Machine Learning, ICML, 2026.

[2] Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness
Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang.
Fortieth AAAI Conference on Artificial Intelligence, AAAI, 2026.

[3] Hessian-Free Online Certified Unlearning
Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei.
Thirteenth International Conference on Learning Representations, ICLR, 2025.

[4] DynFrs: An Efficient Framework for Machine Unlearning in Random Forest
Shurong Wang, Zhuoyang Shen, **Xinbao Qiao**, Tongning Zhang, Meng Zhang.
Thirteenth International Conference on Learning Representations, ICLR, 2025.

[5] Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification
Xinbao Qiao, Wenjing Yan, Ying-Jun Angela Zhang.
Under review.

[6] Illusory Pattern Perception Drives Spurious Inference in Large Language Models
Peihua Mai, Zhuoyan Shao, **Xinbao Qiao**, Meng Zhang, Xinyue Zhou, Yan Pang.
Under review.

[7] Learn What Matters: Data Pruning for Efficient Decentralized Learning
Xinbao Qiao, Xunhao Jiang, Zuozhu Liu, Peng Sun, Meng Zhang.
Under review.